

# The Fermionic Matter Sub-Programme

## Presentation Note 6

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### Abstract

The fermionic matter sub-programme of the Cosmochrony corpus addresses a single central question: where do fermions, chirality, hypercharge, and three generations come from? The answer is that they are not postulated as external fields: they arise as the spinorial face of the admissible Weil module  $V_\rho$ , once its metaplectic Lie-algebraic structure is complexified by the Lorentzian metric established in the geometric branch. The structural chain is:

$$\Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \mathrm{Mp}(2, \mathbb{R}) \Rightarrow \mathrm{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \mathrm{Spin}(3, 1) \simeq \mathrm{SL}(2, \mathbb{C}) \Rightarrow \mathcal{S}_\Pi.$$

Three modular structural results are established in Q14: (A) the admissible spinor bundle  $\mathcal{S}_\Pi$  reproduces the  $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$  electroweak bundle structure from tensor functors of  $\mathcal{S}_\Pi$ ; (B) the projected Dirac operator  $D_{\Pi, g, A}$  contains a canonical projective endomorphism  $E_\Pi$  that enforces the  $V - A$  chiral structure (established via the spinorial BI lift theorem of Q14) and whose  $\gamma_5$ -weighted  $a_4$  coefficient imposes anomaly-cancellation constraints making hypercharge a spectral datum; (C) the saturation invariant  $\sigma_c(n_3) = 3$  has a spinorial multiplicity reading yielding a gauge-singlet three-generation factor  $C_{\mathrm{gen}}^3$ . The colour-coupled quark sector is unconditional at the pointwise level ( $[\mathrm{H}\text{-color}]_{\mathrm{pointwise}}$  established in O31). The structural form of  $E_\Pi$  is now fixed as a Schur complement of the eliminated spinorial block; the open deliverables are its explicit Lorentzian block, the generation-split amplitude (fixed at A4 by matching a discrete chiral-frontier normalisation with the Lorentzian Born–Infeld curvature), and the Yukawa sector.

**Keywords.** fermionic matter; admissible spinor bundle; metaplectic lift; chirality; V-A structure; hypercharge rigidity; anomaly cancellation; three generations; projected Dirac operator; Weil representation; non-injective projection; presentation note.

# Contents

|          |                                                                                                       |           |
|----------|-------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Motivation and Central Question</b>                                                                | <b>3</b>  |
| <b>2</b> | <b>Position in the Cosmochrony Programme</b>                                                          | <b>3</b>  |
| <b>3</b> | <b>Logical Chain of the Sub-Programme</b>                                                             | <b>4</b>  |
| <b>4</b> | <b>Constituent Papers</b>                                                                             | <b>4</b>  |
| 4.1      | Q14: Fermionic matter and chirality from projective Dirac admissibility . . . . .                     | 4         |
| 4.2      | Angular Amplitude Reduction: the $J_3$ split as oriented symplectic area [4] . . . .                  | 5         |
| 4.3      | Q11 Oriented Frontier: recursive diagnostic and resolution of the operator obstruction [6] . . . . .  | 6         |
| 4.4      | Projective Residue as a Schur Complement: localisation and finite/Lorentzian separation [7] . . . . . | 6         |
| <b>5</b> | <b>Inputs from Upstream Results</b>                                                                   | <b>6</b>  |
| <b>6</b> | <b>Outputs to Downstream Branches</b>                                                                 | <b>7</b>  |
| <b>7</b> | <b>Status of Results</b>                                                                              | <b>7</b>  |
| 7.1      | Structural results (unconditional) . . . . .                                                          | 7         |
| 7.2      | Further structural results (unconditional) . . . . .                                                  | 8         |
| 7.3      | Open problems . . . . .                                                                               | 8         |
| <b>8</b> | <b>Open Deliverables</b>                                                                              | <b>8</b>  |
| <b>9</b> | <b>Conclusion</b>                                                                                     | <b>10</b> |

# 1 Motivation and Central Question

The fermionic matter sub-programme answers the following question.

*Central question.* The bosonic spectral stratification — gravity at  $a_2$ , Yang–Mills at  $a_4$  — derives the bosonic sector of the Standard Model from the same admissible spectral functional. Fermions, chirality, hypercharge assignments, and the three-generation structure are not covered by the bosonic synthesis. *Do these fermionic structures also arise as forced consequences of the admissible Weil fibre, or must they be introduced as additional postulates?*

The answer established in Q14 [2] is: they are forced. Fermions arise as the spinorial face of the Weil module  $V_\rho$  after Lorentzian complexification of its metaplectic lift. Chirality and the  $V - A$  structure arise from the projective endomorphism  $E_\Pi$  of the projected Dirac operator, via the spinorial lift of BI parity established in Q14. Hypercharge is not a free parameter: it is constrained by spectral anomaly cancellation. Three generations arise from the spinorial multiplicity reading of the same rank-three invariant  $\sigma_c(n_3) = 3$  that yields the three spatial directions in the geometric branch.

This sub-programme is the first paper of the Q-series that depends simultaneously on all four layers of the programme: the admissible Weil fibre structure (Notes 1 and 5), the closed Lorentzian geometry (Note 2), the gauge bundle (Note 3), and the gauge–gravity synthesis (Q13 [3], to be summarised in Presentation Note 8).

**Remark 1** (Scope: identification only, not mass spectrum). This sub-programme closes the *identification* of the fermionic sector: spinor bundle, chirality, hypercharge quantum numbers, and generation factor. It does not derive the mass spectrum, the Yukawa couplings, or the CKM/PMNS mixing matrices. Those require the explicit construction of  $E_\Pi$  as an operator on the spectral space, and the integration of the projected Yukawa Lagrangian  $\mathcal{L}_Y$ , both of which are listed as open deliverables.

**Remark 2** (Vocabulary note). Throughout this note, *admissibility constraint* replaces the earlier “relaxation” vocabulary. The substrate  $\chi$  is static; fermions are not excitations of  $\chi$  but the spinorial interpretation of the admissible fibre after geometric complexification.

## 2 Position in the Cosmochrony Programme

The fermionic matter sub-programme sits at the apex of Branch III. It depends on all five preceding sub-programmes and provides the fermionic matter content consumed by any downstream phenomenological application.

Inputs from upstream:

- $F_n \simeq V_\rho \simeq L^2(\mathbb{Z}/q\mathbb{Z})$ , canonical symplectic structure of  $V_\rho$ , and the parity involution  $c \leftrightarrow q - c$  (Notes 1 and 5).
- Effective Lorentzian metric  $g^{\mu\nu} = 2\eta^{\mu\nu}$  and effective base manifold  $M_{\text{eff}} = \mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$  (Note 2, Q5b–Q11).
- Admissible principal bundle  $P_{G_\Pi}(M_{\text{eff}}, G_\Pi)$  and gauge connection  $A$  (Note 3, Q6a).
- $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$  gauge group (Note 3).
- Spectral entropy functional  $S_\Pi[g, A]$  and the  $a_4$  variation (Note 4 [8]/Q12, Q13 [3]).
- $H_{\text{eff}} = \text{Sym}^2(V_\rho)$ , three spatial directions from  $\text{Im } \mathbb{H}$ ,  $\sigma_c(n_3) = 3$  (Note 2, Q7 [9]).

### 3 Logical Chain of the Sub-Programme

The sub-programme is organised around the following structural chain.

$$\begin{aligned} \Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \text{Mp}(2, \mathbb{R}) \Rightarrow \text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \text{Spin}(3, 1) \simeq \text{SL}(2, \mathbb{C}) \Rightarrow \mathcal{S}_\Pi \quad (1) \\ \Rightarrow \underbrace{\text{Sym}^2(\mathcal{S}_\Pi) \Rightarrow \text{ad}_{\mathbb{C}}(P_\Pi)}_{\text{SU}(2)_L}, \quad \underbrace{\wedge^2(\mathcal{S}_\Pi) = L_Y}_{\text{U}(1)_Y}, \quad \underbrace{E_\Pi \text{ left-admissible}}_{V-A}, \quad \underbrace{\sigma_c(n_3) = 3 \rightarrow C_{\text{gen}}^3}_{3 \text{ generations}}. \end{aligned}$$

Three distinct mechanisms contribute.

1. *Lorentzian complexification forces the spin group.* The Weil module  $V_\rho$  carries a canonical symplectic structure; its automorphisms lift to  $\text{Mp}(2, \mathbb{R})$ . The effective metric  $g^{\mu\nu} = 2\eta^{\mu\nu}$  (Lorentzian) forces the Lie algebra complexification  $\text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$ , whose Lorentzian integration is  $\text{Spin}(3, 1) \simeq \text{SL}(2, \mathbb{C})$ . The spinor bundle  $\mathcal{S}_\Pi$  is the associated fundamental bundle.
2. *Tensor functors of  $\mathcal{S}_\Pi$  reproduce the electroweak bundle.*  $\text{Sym}^2(\mathbb{C}^2) \simeq \mathfrak{sl}_2(\mathbb{C})$ : the symmetric sector gives the complex adjoint, from which the compact real form selects the  $\text{SU}(2)_L$  gauge sector.  $\wedge^2(\mathbb{C}^2) \simeq \det(\mathbb{C}^2)$ : the determinant line  $L_Y$  supports the  $\text{U}(1)_Y$  factor. No independent electroweak matter bundle is introduced.
3. *The projective endomorphism  $E_\Pi$  enforces chirality and hypercharge.* The projected Dirac operator satisfies the Lichnerowicz-type identity

$$D_{\Pi, g, A}^2 = -(\nabla^{S, A})^2 + \frac{R}{4} + \frac{1}{2}\gamma^\mu \gamma^\nu F_{\mu\nu}^a T_R^a + E_\Pi.$$

By the spinorial lift of BI parity, established in Q14 [2],  $E_\Pi$  is left-admissible ( $P_R E_\Pi P_R = 0$ ), giving the  $V - A$  structure.  $\text{U}(1)_Y$  invariance of  $S_\Pi[g, A, D_{\Pi, g, A}]$  imposes the anomaly-cancellation trace constraint  $\text{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0$ , fixing hypercharge as a spectral coherence datum.

### 4 Constituent Papers

The sub-programme has a single primary paper.

#### 4.1 Q14: Fermionic matter and chirality from projective Dirac admissibility

**Q14** [2] is the sole constituent paper. It is organised around three independent but mutually compatible structural results.

*Theorem A: Spinorial electroweak bundle* (structural; unconditional). The admissible Weil module  $V_\rho$  induces, via the metaplectic lift and the Lorentzian complexification forced by  $g^{\mu\nu} = 2\eta^{\mu\nu}$ , an admissible spinor bundle  $\mathcal{S}_\Pi$ . Its tensor sectors satisfy:

$$\text{Sym}^2(\mathcal{S}_\Pi) \simeq \text{ad}_{\mathbb{C}}(P_\Pi), \quad \wedge^2(\mathcal{S}_\Pi) = L_Y.$$

After selection of the compact real form, these define the  $\text{SU}(2)_L \times \text{U}(1)_Y$  electroweak bundle data without any additional input.

*Theorem B: Chiral selection and hypercharge rigidity* (structural; unconditional). The projected Dirac operator  $D_{\Pi, g, A}$  has the Lichnerowicz-type square above. By the spinorial lift of BI parity, established in Q14 (the antilinear  $J_\Pi = HK$  maps right-chiral modes to non-admissible

directions),  $E_\Pi$  is left-admissible:  $P_R E_\Pi P_R = 0$ . This is the  $V - A$  structure.  $U(1)_Y$  invariance of the projected spectral functional then imposes, at the  $a_4$  level:

$$\text{Tr}_{S_\Pi}(\gamma_5 Y A_\Pi(x)) = 0 \quad \forall x \in M_{\text{eff}},$$

where  $A_\Pi$  is the  $\gamma_5$ -weighted  $a_4$  Seeley–DeWitt density of  $D_{\Pi,g,A}^2$ . This constraint is not postulated: it is the chiral consistency condition of the same  $a_4$  level that yields Yang–Mills dynamics in the bosonic variation. The hypercharge weights  $Y_R$  are thereby not free parameters but spectral coherence data.

*Theorem C: Generation multiplicity* (structural; unconditional). The saturation invariant  $\sigma_c(n_3) = 3$  has two functorially distinct readings of the same rank-three admissible invariant. In the geometric branch (Note 2), it yields three effective spatial directions. In the spinorial branch, the spinorial multiplicity functor maps it to a gauge-singlet generation space:

$$C_{\text{gen}}^3 \subset \ker(\text{ad}_{\text{SU}(2)} \oplus Y).$$

The identity  $\mathbb{R}_{\text{space}}^3 \neq C_{\text{gen}}^3$  is established explicitly (Q14 §5): the spatial and generation spaces are distinct functorial images of the same admissible invariant. Three generations are forced as a gauge-singlet factor. A qualitative mechanism for their splitting is identified in Q14 §6 (summarised below); the explicit  $E_\Pi$  and the amplitude of the splitting remain open.

*Dynamic generation lifting* (localisation and A4 matching). The internal structure of  $C_{\text{gen}}^3$  is probed by the spectrum of  $E_\Pi^2$  (Q14 §6). A static  $J_\Pi$ -real, weight-preserving restriction of  $E_\Pi^2$  has a degenerate outer pair: the two light generations cannot be split statically (static obstruction). The admissible lifts removing this degeneracy form a two-dimensional  $J_\Pi$ -odd sector, the real-split direction being the weight generator  $J_3$  and the mixing direction a doublet rotation. The generation split is not a finite-front observable in the symmetric sense: the raw finite front remains  $J_\Pi$ -equivariant and gives  $u_{\text{fin}} = 0$  [7]. After Lorentzian complexification, however, the chiral projector makes it meaningful to read a  $J_\Pi$ -odd frontier cocycle, and the residual automorphism is the central Weil lift  $R_b = W(-I) = \text{FT}^2$ , which commutes with  $\gamma_5$ : it is blind to chirality and does not obstruct the  $V - A$  selection [6]. The remaining amplitude is fixed at A4 by matching the discrete chiral-frontier normalisation  $\mathcal{N}_A$  [4] with the Lorentzian Born–Infeld curvature  $\mu_\chi^2 = \partial_s^2 \mathcal{B}(0)$  [7], neither ingredient reducing to the other. Generation mixing further requires the complex metaplectic phase.

*Colour sector* (unconditional at the pointwise level). The colour-coupled quark sector  $S_\Pi \otimes V_{\text{color}}$  is obtained by tensoring the admissible spinor bundle with the colour module  $V_{\text{color}}$ . The status of this sector inherits the status of [H-color] from Notes 1 and 3: established unconditionally on the colour-adapted graph, and on the standard graph at the rank, averaged, and effective-exponent levels, with [H-color]<sub>pointwise</sub> proved analytically (O31, exact equality of BFS capacity profiles at finite  $q$ ). The only residual component is the extension beyond the standard Cayley graph to general generating sets, which is non-blocking for the quark bundle.

*Status of the spinorial BI lift.* The spinorial lift of BI parity to the spinor bundle  $S_\Pi$  is established as a theorem in Q14: the BI parity involution  $c \leftrightarrow q - c$ , proved at the scalar Weil level in O18 and BornInfeld, lifts to the antilinear endomorphism  $J_\Pi = HK$  satisfying  $J_\Pi^2 = -1$  and  $J_\Pi \gamma_5 = -\gamma_5 J_\Pi$ , dressing the parity by the weight grading  $H = 2J_3$ . Chirality and  $V - A$  in Q14 are therefore unconditional, their only residual dependence being the axiomatic chain A1–A4 [1] and the upstream BI parity datum (O30, U1).

## 4.2 Angular Amplitude Reduction: the $J_3$ split as oriented symplectic area [4]

This note fixes the numerical observable for the absolute normalisation of the angular generation split before any Weil–BFS computation. A Baker–Campbell–Hausdorff reduction identifies the per-step  $J_\Pi$ -odd  $J_3$  component of the metaplectic cascade with the oriented symplectic area swept per step — the central, Carnot-degree-2 increment of  $\text{Heis}_3$  — and not with a free shear

Table 1: Summary of Q14 results by theorem. Status: P = proved; S = structural.

| Result                  | Central output                                                   | Status                 |
|-------------------------|------------------------------------------------------------------|------------------------|
| Theorem A               | Spinor bundle $\mathcal{S}_\Pi$ ; EW bundle from tensor functors | S (uncond.)            |
| Theorem B ( $V - A$ )   | $P_R E_\Pi P_R = 0$ ; $V - A$ structure                          | S (uncond.)            |
| Theorem B (hypercharge) | $Y_R$ from anomaly-cancellation trace; not free                  | S (uncond.)            |
| Theorem C               | $C_{\text{gen}}^3$ ; three generations as gauge-singlet factor   | S (uncond.)            |
| Colour sector           | $\mathcal{S}_\Pi \otimes V_{\text{color}}$ ; quark bundle        | S (uncond., pointwise) |

parameter. The quantity to measure is the accumulated oriented  $J_3$ -odd area along the cascade, normalised by the projected capacity. A direct evaluation on symmetric BFS shells returns  $\Theta_{\text{raw}} = 0$  identically: the symmetric capacity data fix the radial sector but cannot select an oriented angular branch, so the orientation prescription is an irreducible input prior to the normalisation  $\mathcal{N}_A$ .

### 4.3 Q11 Oriented Frontier: recursive diagnostic and resolution of the operator obstruction [6]

The symmetric-shell observable vanishes ( $\Theta_{\text{raw}} = 0$ ) because the involution  $g \mapsto g^{-1}$  pairs opposite central phases; this note defines the honest replacement, a frontier transfer observable on the directed edges  $g \rightarrow gs$  whose central increment is the Heisenberg cocycle  $a(g) s_b$ , the discrete counterpart of  $\frac{1}{2}[X, Y]$  and hence of the dynamical  $J_3$ . Its main result resolves the operator obstruction: the residual reflection  $R_b = \varphi \circ J_\Pi^{-1} = W(-I) = \text{FT}^2$  is the central Weil lift, and  $[\gamma_5, R_b] = 0$ , so  $R_b$  is blind to chirality and the  $V - A$  selector removes the  $\varphi$ -paired obstruction without a spacetime parity. What remains is the amplitude normalisation  $\mathcal{N}_A$ , not the operator question.

### 4.4 Projective Residue as a Schur Complement: localisation and finite/Lorentzian separation [7]

This note fixes the structural status of the split  $u$  before the explicit  $E_\Pi$ . The projected Dirac square has the Schur–Feshbach form  $E_\Pi = -M^\dagger M$  with  $M = (1 - P) D \Pi_S^*$ , exhibiting  $E_\Pi$  as the Schur complement of the eliminated spinorial block;  $u$  is localised in the  $J_\Pi$ -odd part of  $E_\Pi^2|_{C_{\text{gen}}^3}$  as the antiunitary equivariance defect  $\Delta_\chi(P) = \pi_{LL} - \tau \overline{\pi_{RR}} \tau^{-1}$ . The finite locking is  $J_\Pi$ -equivariant, so  $u_{\text{fin}} = 0$  (finite/Lorentzian separation): the split cannot originate in the finite cascade and is a Lorentzian datum, its magnitude reduced to the Born–Infeld saturation curvature  $\mu_\chi^2 = \partial_s^2 \mathcal{B}(0)$ . The Schur transversality on which the A4 matching was conditional is reduced, in the same note, to the non-vanishing of the projected commutator  $[\partial_s P|_0, J_\Pi]|_{\text{span}(e_+, e_-)}$ , with an exhaustive transverse/null dichotomy including the zero-mode subcase. The determinantal Born–Infeld margin  $\mathcal{B}(s)$ , the antisymmetric structure of the chiral modulus, and the reduction of  $\mu_\chi^2$  to the Lorentzian genus of the chiral polarisation are developed in the companion note [5].

## 5 Inputs from Upstream Results

1.  $F_n \simeq V_\rho$  and symplectic structure (Notes 1, 5): the identification of the admissible fibre as the Weil representation is the starting point of the metaplectic lift. Without this, the spinor bundle  $\mathcal{S}_\Pi$  has no canonical definition.
2. Lorentzian metric  $g^{\mu\nu} = 2\eta^{\mu\nu}$  (Note 2, Q5b–Q11): the Lorentzian signature forces the complexification  $\text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$ . Without the Lorentzian metric, the spin group would

remain  $\mathrm{Sp}(2, \mathbb{R})$  and no Dirac spinors would arise.

3. **Gauge bundle and  $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \times \mathrm{SU}(3)$**  (Note 3, Q6a): the admissible principal bundle and gauge connection  $A$  enter the spin–gauge connection  $\nabla^{S,A}$  of the projected Dirac operator.
4.  **$a_4$  Yang–Mills variation** (Q12 [10], Q13 [3]): the spectral anomaly cancellation of Theorem B uses the same  $a_4$ -level structure that yields Yang–Mills dynamics in the bosonic sector. The fermionic  $a_4$  coefficient is a structural extension of the bosonic one. The gauge–gravity synthesis of Q13 is the scientific dependency here; its presentation-level treatment will appear in Presentation Note 8.
5.  $\sigma_c(n_3) = 3$  (Note 1, O23): the rank-three saturation invariant provides the spinorial multiplicity input for Theorem C.
6. **BI parity involution** (BornInfeld, O18): the scalar BI parity  $c \leftrightarrow q - c$  is the input from which the spinorial BI lift is derived; its derivation at the scalar level is complete.

## 6 Outputs to Downstream Branches

Table 2: Principal outputs and downstream consumers.

| Output                                                         | Consumer                | Status                 |
|----------------------------------------------------------------|-------------------------|------------------------|
| Admissible spinor bundle $\mathcal{S}_\Pi$                     | Phenomenology, Q-series | S                      |
| $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ from tensor functors | SM phenomenology        | S                      |
| $V - A$ chiral structure                                       | Weak interaction pheno  | S (uncond.)            |
| Hypercharge as spectral datum                                  | SM charge assignment    | S (uncond.)            |
| Three-generation factor $C_{\mathrm{gen}}^3$                   | Mass/mixing notes       | S                      |
| Quark bundle $\mathcal{S}_\Pi \otimes V_{\mathrm{color}}$      | QCD sector              | S (uncond., pointwise) |
| $E_\Pi$ as projective endomorphism                             | Yukawa sector (open)    | S (open)               |

## 7 Status of Results

### 7.1 Structural results (unconditional)

1. *Admissible spinor bundle* (Theorem A):  $\mathcal{S}_\Pi$  is induced from  $V_\rho$  via the metaplectic lift and Lorentzian complexification. The construction is algebraic and functorial; it depends on the Lorentzian closure of Q5b–Q11 but not on the spinorial BI lift.
2.  *$\mathrm{SU}(2)_L$  from  $\mathrm{Sym}^2(\mathcal{S}_\Pi)$*  (Theorem A): the symmetric tensor sector gives the complex adjoint  $\mathrm{ad}_{\mathbb{C}}(P_\Pi) \simeq \mathfrak{sl}_2(\mathbb{C})$ ; after compact real-form selection, the  $\mathrm{SU}(2)_L$  gauge sector is identified.
3.  *$\mathrm{U}(1)_Y$  from  $\wedge^2(\mathcal{S}_\Pi)$*  (Theorem A): the determinant line  $L_Y = \wedge^2(\mathcal{S}_\Pi)$  is the geometric support of the hypercharge factor.
4. *Generation multiplicity* (Theorem C):  $C_{\mathrm{gen}}^3 \subset \ker(\mathrm{ad}_{\mathrm{SU}(2)} \oplus Y)$  is a gauge-singlet three-generation space arising from the spinorial multiplicity reading of  $\sigma_c(n_3) = 3$ . The spatial and generation readings of the rank-three invariant are distinct ( $\mathbb{R}_{\mathrm{space}}^3 \neq C_{\mathrm{gen}}^3$ ; Q14 §5).

## 7.2 Further structural results (unconditional)

1. *Left-admissibility of  $E_\Pi$*  (Theorem B):  $P_R E_\Pi P_R = 0$  follows from the spinorial BI lift (Q14), now established.
2.  *$V - A$  chiral structure* (Theorem B): the condition  $P_R E_\Pi P_R = 0$  selects the left-chiral admissible projection  $P_L \mathcal{S}_\Pi$ , giving the  $V - A$  coupling of weak interactions.
3. *Hypercharge rigidity* (Theorem B): the anomaly-cancellation trace condition  $\text{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0$  constrains  $Y_R$  to spectral coherence data; hypercharge is not a free parameter of the framework.
4. *Quark bundle* (§6):  $\mathcal{S}_\Pi \otimes V_{\text{color}}$  is structural and unconditional at the pointwise level: the spinorial BI lift is established (Q14) and  $[\text{H-color}]_{\text{pointwise}}$  is proved analytically (O31).
5. *A4 Schur-transversality reduction* [7]: the conditional residual of the A4 matching is reduced to the non-vanishing of the projected commutator  $[\partial_s P|_0, J_\Pi]_{\text{span}(e_+, e_-)}$ , with an exhaustive transverse/null dichotomy including the zero-mode subcase; the transverse branch determines the split via the electric genus of the chiral polarisation [5], the Schur-null branch is marginal.

## 7.3 Open problems

1. *Explicit form of  $E_\Pi$* .  $E_\Pi$  is identified as the zero-order endomorphism of the projected Dirac square, the “internal spectral residue of non-injective projection.” Its structural form is now fixed as the Schur complement of the eliminated spinorial block,  $E_\Pi = -M^\dagger M$  with  $M = (1 - P)D \Pi_S^*$  [7]; what is not yet constructed is its explicit Lorentzian block  $1 - P(s)$  along the  $J_\Pi$ -odd modulus. The inter-generation splitting is localised in the  $J_\Pi$ -odd part of  $E_\Pi^2$ , with amplitude fixed at A4 by matching a discrete chiral-frontier normalisation with the Lorentzian Born–Infeld curvature; without the explicit Lorentzian  $E_\Pi$ , the three-generation mass spectrum and the Yukawa sector cannot be derived.
2. *Normalisation of  $\mathcal{L}_Y$* . The Yukawa Lagrangian  $\mathcal{L}_Y$  is identified as arising from the trace  $\text{Tr}_{\mathcal{S}_\Pi}[E_\Pi^2]$  variation. Its integral normalisation and the derivation of the mass matrix from the spectral data of  $E_\Pi$  are open.
3.  $[\text{H-color}]_{\text{pointwise}}$  *beyond the standard graph*. Inherited from Notes 1 and 3.  $[\text{H-color}]_{\text{pointwise}}$  is proved on the standard Cayley graph (O31); its extension to general generating sets is the only residual, and is non-blocking for the quark bundle, which is already established at the pointwise level.

## 8 Open Deliverables

The derivation of the spinorial BI lift, formerly the first open deliverable, is now established as a theorem in Q14 [2]: the antilinear lift  $J_\Pi = HK$  dresses the BI parity by the weight grading, making Theorems B and C unconditional. Two open deliverables remain.

**Open deliverable 1: Explicit  $E_\Pi$  and the Yukawa sector.** The mass spectrum of charged leptons and quarks, and the CKM/PMNS mixing matrices, require the explicit spectral construction of  $E_\Pi$  as an operator on the admissible Hilbert space. This requires integrating the projected Yukawa Lagrangian  $\mathcal{L}_Y$  and extracting the mass matrix from the eigenvalues of  $E_\Pi^2$ . This is the primary open problem of the fermionic sector. The structural form of  $E_\Pi$  is now fixed as the Schur complement of the eliminated spinorial block [7]; the open part is its explicit Lorentzian block  $1 - P(s)$  along the  $J_\Pi$ -odd modulus. The symmetric finite front is  $J_\Pi$ -equivariant and gives  $u_{\text{fin}} = 0$ , but the discrete chiral-frontier normalisation  $\mathcal{N}_A$  survives the



operator obstruction, since the residual reflection  $R_b = \text{FT}^2$  is blind to chirality ( $[\gamma_5, R_b] = 0$ ) [6]. The split amplitude is therefore fixed at A4 by matching this discrete normalisation with the Lorentzian Born–Infeld curvature  $\mu_\chi^2 = \partial_s^2 \mathcal{B}(0)$ , neither ingredient reducing to the other:

$$|u| = \text{A4Match}(\mathcal{N}_A, \mu_\chi^2).$$

The operator side of this matching is no longer conditional: by [7] the Schur transversality is reduced to the non-vanishing of the projected commutator  $[\partial_s P|_0, J_\Pi]_{\text{span}(e_+, e_-)}$ , with an exhaustive transverse/null dichotomy. In the Schur-transverse branch the A4 chiral polarisation is electric, hence  $\mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu} < 0$ ,  $\mu_\chi^2 < 0$ , and the split opens spontaneously [5]; the genus is determined. What remains open is the amplitude  $|u|$  (this A4 matching) and the Yukawa sector downstream of  $u$ . The amplitude is reduced in the A4 companion to the sign of the cubic backreaction  $\mathcal{P} \cdot \mathcal{R}_\chi$  and to the boundary-versus-interior selection of the Born–Infeld saturation problem. What remains quantitative is therefore this A4 matching, the integral normalisation of  $\mathcal{L}_Y$ , and generation mixing via the complex metaplectic phase.

**Open deliverable 2: Full matter content in Lorentzian signature.** The complete fermionic content  $S_\Pi^{\text{matter}} = (\mathcal{S}_\Pi \oplus (\mathcal{S}_\Pi \otimes V_{\text{color}})) \otimes C_{\text{gen}}^3$  in full Lorentzian signature, with explicit gauge couplings and three-generation Yukawa structure, is the downstream target. It requires the resolution of Deliverable 1 above, the colour sector being already established at the pointwise level (O31).

## 9 Conclusion

The fermionic matter sub-programme establishes that fermions, chirality, hypercharge, and three generations are not external additions to the Cosmochrony framework. They are the spinorial face of the admissible Weil module  $V_\rho$ , revealed through the Lorentzian complexification of its metaplectic structure.

The electroweak bundle  $SU(2)_L \times U(1)_Y$  arises from the tensor functors of the admissible spinor bundle  $\mathcal{S}_\Pi$  (Theorem A, unconditional). The  $V - A$  chiral structure and hypercharge rigidity arise from the projective endomorphism  $E_\Pi$  (Theorem B, unconditional). Three generations arise from the spinorial multiplicity reading of  $\sigma_c(n_3) = 3$  (Theorem C, unconditional). The colour-coupled quark sector follows by tensoring with  $V_{\text{color}}$ , unconditional at the pointwise level (O31).

The inter-generation splitting is localised: statically obstructed, living in a two-dimensional  $J_\Pi$ -odd lift sector, and vanishing on the symmetric finite front ( $u_{\text{fin}} = 0$ ,  $J_\Pi$ -equivariant). Its operator obstruction is resolved, the residual reflection  $R_b = \text{FT}^2$  being blind to chirality ( $[\gamma_5, R_b] = 0$ ). The remaining open components are the explicit Lorentzian  $E_\Pi$ , the split amplitude fixed at A4 by matching the discrete chiral-frontier normalisation with the Lorentzian Born–Infeld curvature  $\mu_\chi^2$ , and the derivation of the Yukawa sector and mass spectrum.

## References

- [1] J. Beau. Admissible Non-Injective Transitions as the Primitive of Physical Description, 2026. Cosmochrony Foundation Paper "M", doi:10.5281/zenodo.19616329.
- [2] Jérôme Beau. Fermionic Matter and Chirality from Projective Dirac Admissibility, 2026. "Q14" paper, doi:10.5281/zenodo.20218409.
- [3] Jérôme Beau. Gauge–Gravity Spectral Synthesis: Joint Einstein–Yang–Mills Equations, Born–Infeld Completion, and the Hierarchy Ratio from Projective Spectral Entropy, 2026. "Q13" paper, doi:10.5281/zenodo.20209859.
- [4] Jérôme Beau. Reduction of the Angular Amplitude Observable: the  $J_3$  Split as Accumulated Oriented Symplectic Area, 2026. doi:10.5281/zenodo.20601228.
- [5] Jérôme Beau. The Born–Infeld Saturation Margin of the Chiral Modulus: Antisymmetric Structure and the Reduction of the Generation Split to a Lorentzian Genus, 2026. Fermionic matter sub-programme companion note. doi:10.5281/zenodo.20633931.
- [6] Jérôme Beau. The Oriented Frontier Observable: a Recursive Q11 Diagnostic for the Angular Generation Split, 2026. doi:10.5281/zenodo.20601245.
- [7] Jérôme Beau. The Projective Residue as a Schur Complement: Reduction of the Generation Split to the Chiral Asymmetry of Projection Locking, 2026. doi:10.5281/zenodo.20601040.
- [8] Jérôme Beau. The Spectral Gravity Sub-Programme, 2026. Presentation Note 4, doi:10.5281/zenodo.20533351.
- [9] Jérôme Beau. Three Admissible Directions and Three Spatial Dimensions: A Structural Bridge Candidate, 2026. "Q7" paper, doi:10.5281/zenodo.19802123.
- [10] Jérôme Beau. Yang–Mills Dynamics from Projective Spectral Entropy: Vertical Heat-Kernel Variation on the Admissible Fibre, 2026. "Q12" paper, doi:10.5281/zenodo.20261952.